



A novel method for functional brain networks based on static cerebral blood flow

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ABSTRACT

Cerebral blood flow (CBF) offers a quantitative and reliable measurement for brain activity and is increasingly used to study functional networks. However, current methods evaluate inter-regional relations mainly based on CBF temporal dynamics, which suffers from low signal-to-noise ratio and poor temporal resolution. Here we proposed a method to construct functional brain networks by estimating shape similarity (index by Jensen–Shannon divergence) in probability distributions of regional static CBF measured by arterial spin labeling perfusion imaging over a scanning period. Based on CBF data of 30 healthy participants from 10 visits, we found that the CBF networks exhibited non-trivial topological features (e.g., small-world organization, modular architecture, and hubs) and showed low-to-fair test-retest reliability and high between-subject consistency. We further found that interregional CBF similarities were depended on anatomical distance and differed between high- and lower-order subnetworks. Moreover, interregional CBF similarities within high-order subnetworks showed significantly lower reliability than those within low-order subnetworks. Finally, we showed that nodal degree of the CBF networks were related to regional sizes and CBF levels and spatially aligned with maps of the dopamine transporter and metabolic glutamate receptor 5 intensities, expression levels of genes primarily enriched in cholesterol-related pathways and endothelial cells, and meta-analytic activations related to memory, language, and executive function. Altogether, our proposed method provide a novel, relatively reliable, and neurobiologically meaningful means to study functional network organization of the human brain.

1. Introduction

Functional connectivity has become an effective method for characterizing intrinsic functional dynamics (Van Dijk et al., 2010) and been applied to assess alterations in various neuropsychiatric disorders, such as major depression disorder (Ye et al., 2015; Zhang et al., 2011) and dementia (Pievani et al., 2011; Wang et al., 2013), associations with different cognitive abilities (Li et al., 2017; Sripada et al., 2021) and intelligence (Langer et al., 2012), and changes during development (Gao et al., 2015; Hwang et al., 2013) and aging (Sala-Llonch et al., 2014). Currently, functional connectivity is estimated mainly based on blood oxygen level-dependent (BOLD) signals measured by functional magnetic resonance imaging (fMRI). However, BOLD signals are an indirect index for neuronal activity, which measure hemodynamic changes

influenced by cerebral blood flow (CBF), blood volume, and blood oxygenation (Hillman, 2014). By contrast, CBF as measured by arterial spin labeling (ASL) perfusion imaging is a direct and quantitative biological index for brain function (Wang et al., 2008; Wong, 2014). Moreover, CBF couples with cerebral metabolic rates for glucose and oxygen consumption (Vaishnavi et al., 2010a). Therefore, CBF has been increasingly used to evaluate functional connectivity in recent years.

To date, several methods have been proposed to estimate functional connectivity based on CBF. For example, an early study applied independent component analysis to CBF time series and identified 5 distinct resting-state networks with spatial and temporal characteristics closely matched those derived from BOLD-based functional networks (De Luca et al., 2006). Subsequently, Liang and colleagues constructed functional networks by calculating Pearson correlation of regional CBF time series

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and found hub regions that largely overlapped with those in BOLD-based functional networks (Liang et al., 2014a). The Pearson correlation-based method was also extended to multiple regression (Li et al., 2020) and cross-correlation (Zou et al., 2009). Despite the success, it should be noted that such CBF time series-based functional connectivity methods are limited by the low signal-to-noise ratio and poor temporal resolution of CBF time series (Alsop et al., 2015; Wong, 2014; Zou et al., 2009). These drawbacks largely limit the application scenarios of the CBF time series-based functional connectivity methods. Accordingly, it is necessary to develop new methods to circumvent these drawbacks for studying functional connectivity or networks using CBF.

Here, we proposed a novel method to construct functional brain networks by estimating shape similarity in probability distributions of regional CBF based on a static mean perfusion map over a scanning period. Previous studies have shown that the static mean perfusion map has high signal-to-noise ratio and exhibits high test-retest (TRT) reliability (Zou et al., 2015). Specifically, we employed a kernel density estimation method to fit probability distributions of regional CBF, between which the shape similarity was quantified based on Jensen–Shannon divergence (JSD). For the resulting CBF networks, we systematically characterized their topological architecture, evaluated their TRT reliability, between-subject consistency, and potential influencing factors, and explored their chemoarchitectonic, genetic, and cognitive correlates. Fig. 1 presents the overall data processing pipeline. In this study, we hypothesized that shape similarity in probability distributions of static CBF between regions could offer a TRT reliable, neurobiologically meaningful, and cognitively relevant means for studying functional brain networks.

2. Materials and methods

2.1. Ethics statement

All datasets used in this study were from public repositories. For each dataset, the subject recruitment processes and informed consent forms were ratified by the corresponding Institutional Review Board. This study was approved by the Institutional Review Board of the Institute for Brain Research and Rehabilitation at South China Normal University.

2.2. Participants and data acquisition

A publicly available TRT dataset was used in this study (Chen et al., 2015). This dataset contained 30 subjects (15 females; mean age 24, SD = 2.41), who were confirmed to have no history of neurological or psychiatric disorders, substance abuse, or head injuries resulting in consciousness loss. Each subject was scanned ten times over one month with one scan every three days. Specifically, ASL perfusion MRI data were acquired using a 3.0-Tesla Discovery MR750 scanner (General Electric, Milwaukee, WI, USA), employing a 3D pseudo-continuous ASL with the following parameters: repetition time (TR) = 4.834 s; echo time (TE) = 11.088 ms; flip angle (FA) = 111°; slice thickness = 3 mm; no gap; slice number = 100; field of view (FOV) = 100% of phase FOV; in-plane matrix = 128 × 128; control-label pair number = 44; total acquisition time = 411 s; labeling duration = 1500 ms; post-labeling delay = 1525 ms. These parameters did not follow the recommended guidelines by Alsop and colleagues (Alsop et al., 2015) since the data were collected earlier than the release of the guidelines. Background suppression and vascular crushing were used during the data collection. In addition, a T1-weighted Fast Spoiled Gradient echo was carried out to acquire a high-resolution anatomical image for each subject: TR = 8.1 ms; TE = 3.1 ms; inversion time = 450 ms; FA = 8°; FOV = 256 × 256 mm²; in-plane matrix = 256 × 256; voxel size = 1.0 × 1.0 × 1.0 mm³;

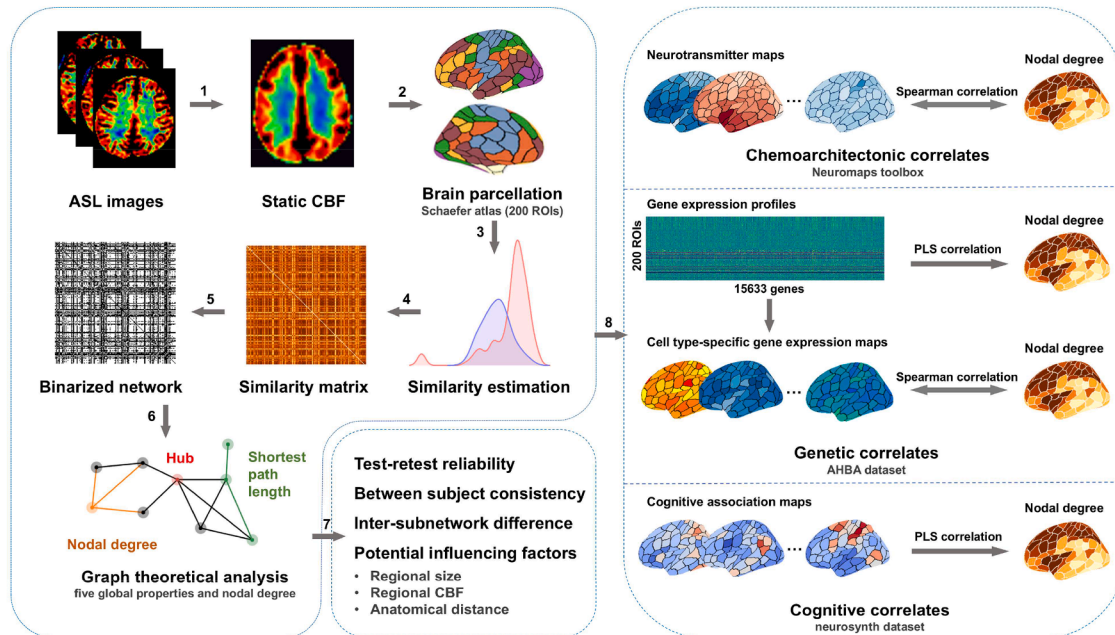


Fig. 1. A flowchart of data analytical pipeline in this study. (1) Individual ASL images were first processed to generate a static CBF map. (2) The CBF map was parceled into 200 ROIs in terms of the Schaefer atlas. (3) For each ROI, a probability distribution was obtained based on CBF values of voxels in the ROI. (4) For each pair of ROIs, the similarity in their probability distributions was estimated based on Jensen–Shannon divergence. (5) A sparsity-based procedure was used to threshold the resultant CBF matrix into a series of binary networks. (6) Graph theory-based network metrics were calculated to characterize the topological organization of the binary networks. (7) For interregional CBF similarity and/or topological properties of CBF networks, analyses were conducted to assess their TRT reliability, between-subject consistency, potential influencing factors, and differences between high- and low-order subnetworks. (8) Chemoarchitectonic, genetic, and cognitive correlates of CBF networks were examined by correlating nodal degrees of the networks with publicly available neurotransmitter, gene expression, and cognitive association maps. ASL, arterial spin labeling; CBF, cerebral blood flow; ROIs, regions of interest; AHBA, Allen Human Brain Atlas.

and 176 sagittal slices). A written informed consent was obtained from each subject before data collection.

2.3. ASL images processing

The ASL images were processed to derive individual quantitative static CBF maps using a GE-AW workstation (ADW 4.7, GE Healthcare) with the READY view post-processing tool based on the following algorithm:

$$CBF = 6000 * \lambda \frac{\left(1 - \exp\left(-\frac{ST}{T_{1t}}\right)\right) \exp\left(\frac{PLD}{T_{1b}}\right)}{2T_{1b}(1 - \exp\left(-\frac{LT}{T_{1b}}\right)) \varepsilon * NEX_{PW}} \left(\frac{PW}{SF_{PW}PD}\right)$$

where T_{1b} and T_{1t} are the longitudinal relaxation of arterial blood (1.6 s) and gray matter (1.2 s), respectively; ST is the saturation time (2 s); λ is the partition coefficient and is set to the whole brain average, 0.9; ε is a combination of both inversion efficiency (0.8) and background suppression efficiency (0.75), resulting in an overall efficiency of 0.6; PLD is the post-labeling delay (1525 ms); LT is the labeling duration (1500 ms); PD is the partial saturation of the reference image, which is corrected using a T_{1t} of 1.2 s; PW refers to perfusion-weighted or original difference images; SF_{PW} is the scaling factor of the PW sequence; NEX_{PW} is the number of excitations for the PW images. The resultant CBF was reported in mL/100g/min units.

Before the CBF quantification, no explicit head motion correction was performed for ASL images. We thus carefully reviewed the CBF maps and excluded any maps showing visible motion artifacts from further analysis. The CBF maps were further corrected for partial volume effects using the ASLtbx toolbox (Hu et al., 2010; Wang et al., 2008), co-registered to corresponding structural images, normalized to the MNI space, resampled to 3-mm isotropic voxels, and spatially smoothed using a 6-mm full-width half-maximum Gaussian kernel. Finally, the CBF maps were mean-scaled to reduce the effects of inter-subject variability in global CBF by dividing each voxel's value by the whole-brain mean value for each subject. Notably, both spatial smoothing and mean scaling were restricted to a gray matter mask derived from the gray matter tissue probability map in the SPM12 toolbox (threshold = 0.4) to rule out the impact of low signal-to-noise ratio in white matter.

2.4. Construction of CBF networks

A network is composed of nodes and edges linking the nodes. In this study, we constructed a functional brain network for each subject based on his or her processed quantitative static CBF map.

Definition of network nodes. Network nodes were defined in terms of the Schaefer's atlas (Schaefer et al., 2018), which is widely used in previous brain network studies for cortical parcellation. This atlas parcels the cerebral cortex into 200 regions of interest (ROIs). Specifically, these ROIs were categorized into seven subnetworks (Thomas Yeo et al., 2011): visual network (VN), somatomotor network (SMN), dorsal attention network (DAN), ventral attention network (VAN), limbic network (LN), fronto-parietal control network (FPN), and default mode network (DMN).

Definition of network edges. Network edges were defined as JSD-based shape similarity (JSDs) in probability distributions of regional CBF between ROIs. We first extracted CBF values within each ROI and estimated its probability density function using MATLAB function, *ksdensity*. The probability density functions were then transformed into probability distributions (PDs). Finally, the JSDs was defined by:

$$JSDs(P, Q) = 1 - \sqrt{JSD(P \parallel Q)}$$

$$JSD(P \parallel Q) = \frac{1}{2} \sum_{i=1}^n P(i) \log \frac{P(i)}{M(i)} + \frac{1}{2} \sum_{i=1}^n Q(i) \log \frac{Q(i)}{M(i)}$$

$$M = \frac{1}{2}(P + Q)$$

where P and Q denote regional PDs, and n denotes the sampling points number during the probability density function estimation. The value range for JSDs is [0, 1], with a larger value indicating a higher similarity. We evaluated the effects of the number of sampling points on regional probability density functions using the same method as our previous study (Wang et al., 2016). Briefly, the Frechet distance was first calculated for each ROI to quantify the similarity of its probability density functions that were estimated using different numbers of sampling points (2^4 vs. 2^5 , 2^5 vs. 2^6 , 2^6 vs. 2^7 , 2^7 vs. 2^8 , 2^8 vs. 2^9 , and 2^9 vs. 2^{10}). This analysis was conducted using the CBF maps obtained from the first scan of the subjects, which resulted in a 30 (participants) $\times$ 200 (ROIs) $\times$ 6 (pairs of adjacent sampling number) Frechet distance matrix. The Frechet distance matrix was subsequently averaged across subjects to derive a group-level Frechet distance matrix (200 $\times$ 6). Finally, nonparametric permutation tests (10,000 times) were used to compare any two adjacent columns in the group-level Frechet distance matrix. We found that the similarity of regional probability density functions significantly and continuously increased with the increasing number of sampling points [$p < 0.05$, corrected by a false discovery rate (FDR) procedure] (Table S1). These findings suggest that more sampling points are associated with more stable estimation of regional probability density functions from CBF. To avoid overfitting and to provide a trade-off between stable estimation and computational cost, 2^8 was chosen in this study, consistent with our previous study (Wang et al., 2016).

2.5. Graph-based topological characterization of CBF networks

We utilized graph-based network measures to characterize the CBF networks constructed above. All network analyses were performed with the GREYNA toolbox (Wang et al., 2015).

Threshold selection. Prior to network measure calculation, a sparsity-based thresholding procedure was used to binarize the CBF networks, which is defined as the ratio of the number of actual edges to the maximum possible number of edges in a network. The sparsity-based thresholding procedure ensures the same number of edges across subjects. Specifically, each CBF network was thresholded at 20 logarithmically spaced points in the sparsity range of [0 1], in line with our previous study (Yin et al., 2023).

Global properties. We calculated five global properties (clustering coefficient, C_p , shortest path length, L_p , local efficiency, E_{loc} , global efficiency, E_{glob} , and modularity, Q) at each sparsity for each binary network. To test whether the CBF networks are non-randomly organized, all global properties were further normalized by dividing them by the corresponding average of 100 matched random networks. The random networks were generated using a topological rewiring algorithm to preserve the same degree distribution as the CBF networks (Maslov and Sneppen, 2002). Typically, a small-world, highly efficient, and modular network should fulfill the following conditions: normalized $C_p > 1$ and normalized $L_p \sim 1$, normalized $E_{loc} > 1$ and normalized $E_{glob} \sim 1$, and normalized $Q > 1$. Considering the dependence of network properties on sparsity, we further calculated the area under the curve (AUC) to provide a summary scalar for each global network measure.

Nodal degree. We calculated nodal degree at each sparsity for each binary network to characterize the roles of individual nodes in the CBF networks. To provide a summary scalar for each node, we also calculated the AUC for nodal degree. The AUCs of nodal degrees were subsequently averaged across subjects and used to fit degree distribution using different models (power law, exponential, and exponentially truncated power law). Hubs were defined as nodes whose nodal degrees (AUC) ranked in the top 10%.

2.6. Evaluation of CBF networks

TRT reliability. We quantified TRT reliability of the CBF networks using intraclass correlation coefficient (ICC) (Shrout and Fleiss, 1979). Specifically, we calculated the ICC for each global network measure (AUC), nodal degree (AUC) of each region, and interregional CBF similarity between each pair of regions. Specifically, for a given measure, the ICC was calculated as:

$$ICC = \frac{MS_R - MS_W}{MS_R + (k - 1)MS_W}$$

where MS_R represents the mean square of between-subject variance, MS_W represents the mean square of within-subject variance, and k represents the number of repeated observations per subject (here, $k = 10$). Similar to our previous studies (Wang et al., 2011, 2016; Li et al., 2021b; Yin et al., 2023), the TRT reliability was categorized as poor ($ICC < 0.25$), low ($0.25 < ICC < 0.4$), fair ($0.4 < ICC < 0.6$), good ($0.6 < ICC < 0.75$), and excellent ($0.75 < ICC < 1$).

Between-subject consistency. To assess between-subject consistency of the CBF networks, we calculated Pearson correlation between each pair of subjects in their nodal degrees (across nodes) and CBF similarity matrices (across edges) averaged across all scans.

Inter-subnetwork differences. To study whether the CBF networks were related to functional topography of the cortex, we examined the differences between low-order (VN and SMN) and high-order (DAN, VAN, FPN, and DMN) subnetworks in nodal degrees and intra-subnetwork interregional CBF similarities averaged across all subjects and scans using two-sample t -tests. Significance levels of the t statistics were estimated through nonparametric permutation tests (10,000 times). Similarly, we examined the differences between the low- and high-order subnetworks in the TRT reliability of nodal degrees and intra-subnetwork interregional CBF similarities.

Potential influencing factors. For nodal degrees averaged across all subjects and scans, we examined their relationships with regional sizes and mean CBF levels using Spearman correlation (across nodes). Significance levels of these correlations were estimated with spin tests (Alexander-Bloch et al., 2018; Váša et al., 2018). Spin test is a spatial permutation approach based on angular permutations of spherical projections at the cortical surface. It preserves the spatial covariance structure of the data and thus corrects for potential confounding effects of spatial autocorrelation. Accordingly, spin test is widely used to assess the statistical significance of the correlation between two cortical maps (i.e., source and target maps). Specifically, we first generated 10,000 random spatial rotations (i.e., spins) of the mean nodal degree map (i.e., source map), each of which was correlated with the target map (i.e., regional size map or mean CBF map). Then, a P -value was calculated as the proportion of spins for which the absolute values of the resultant correlation coefficients exceeded or equalled the actually observed absolute value of the correlation coefficient between the source and target maps. In addition, we examined the relationship between interregional CBF similarities averaged across all subjects and scans and interregional anatomical distances (Euclidean distance) using Spearman correlation (across edges).

2.7. Chemoarchitectonic correlates of CBF networks

To investigate the chemoarchitectonic correlates of the CBF networks, we first obtained 38 neurotransmitter intensity maps of healthy subjects from https://github.com/netneurolab/hansen_receptors/tree/main/data/PET_parcellated (Hansen et al., 2022). These maps included 5-hydroxytryptamine receptor 1a (5-HT_{1A}), 5-HT_{1B}, 5-HT_{2A}, 5-HT₄, 5-HT₆, 5-HT₇, alpha-4 beta-2 nicotinic receptor ($\alpha 4 \beta 2$), cannabinoid receptor 1 (CB₁), dopamine receptor 1 (D₁), D₂, dopamine transporter (DAT), γ -Aminobutyric acid type A (GABA_A), Histamine H₃ receptor (H₃), muscarinic 1 (M₁), metabolic glutamate receptor 5

(mGluR₅), opioid receptor μ (MOR), noradrenaline transporter (NET), N-methyl-D-aspartate (NMDA), and vesicular acetylcholine transporter (VACHT). The D₂ intensity map using raclopride tracers was excluded from further analysis due to the unreliable binding of raclopride tracers in the cortex. Details on these neurotransmitter intensity maps can be found at <https://github.com/netneurolab/neuromaps/wiki/Annotation-information> (Markello et al., 2022). We then analyzed the relationship between nodal degrees averaged across all subjects and scans and regional mean intensities of each neurotransmitter using Spearman correlation (across nodes). Significance levels of these correlations were evaluated using spin-based permutation tests (10,000 times). The FDR procedure was employed to correct for multiple comparisons across all correlations at the level of $q < 0.05$.

2.8. Genetic correlates of CBF networks

To investigate the genetic correlates of the CBF networks, we examined the relationship between nodal degrees averaged across all subjects and scans and transcriptional profiles from the AHBA dataset. For genes that strongly contributed to the relationship, we further performed gene ontology (GO) enrichment analysis to identify their related biological process.

AHBA dataset. The AHBA dataset is a publicly available online resource, which contains brain-wide transcriptional information from six healthy adult human donors (age, 24 - 57 years; 5 males and 1 female) with no known neuropathological or neuropsychiatric disease history (Hawrylycz et al., 2012). Specifically, the transcriptional activity was recorded for 20,737 genes from 3702 spatially distinct tissue samples that covered almost the entire brain. The tissue samples were collected from the left hemisphere for 4 donors and both hemispheres for 2 donors. For more details, see <http://human.brain-map.org/>. Standardized workflows (Arnatkeviciute et al., 2019) were used to preprocess the gene data (see Supplementary Methods for details).

Relationship between CBF networks and transcriptional profiles. To investigate the relationship between the CBF networks and transcriptional profiles, we performed the partial least squares (PLS) regression to predict nodal degrees averaged across all subjects and scans with regional expression levels of all genes. The first component of the PLS (PLS1) was the linear combination of regional expression levels of all genes that exhibited the strongest correlation with the nodal degrees. Significance level of the correlation was estimated by re-running the PLS regression for nodal degrees simulated via a spin-based permutation test (10,000 times) to correct for spatial autocorrelation. If the correlation was significant, the weights of all genes to form the PLS1 were further Z-transformed, and genes with an absolute Z-score > 1.64 were considered to contribute to the correlation strongly. For the identified genes that strongly contributed to the PLS1, we further performed gene ontology (GO) enrichment analysis to search for their related GO terms (Supplementary Methods). Of note, the GO enrichment analysis was performed separately for genes showing the strongest positive and negative contributions to the PLS1 (i.e., PLS1+ and PLS1- genes). Furthermore, to investigate whether nodal degrees of the CBF networks are related to cellular architecture-specific gene expression, we calculated Spearman correlation between nodal degrees and mean expression levels of genes enriched in various cell classes, including excitatory neurons, inhibitory neurons, oligodendrocyte progenitor cells, astrocytes, endothelial cells, microglia, and oligodendrocytes (Arnatkeviciute et al., 2021). Statistical levels of all correlations were estimated using spin-based permutation tests (10,000 times), followed by multiple comparison correction across all cell classes using the FDR procedure ($q < 0.05$).

2.9. Cognitive correlates of CBF networks

To examine the cognitive correlates of the CBF networks, we correlated nodal degrees averaged across all subjects and scans with multiple meta-analytic activation maps using PLS regression. The activation

maps were derived from the NeuroSynth database (Yarkoni et al., 2011) (<https://github.com/neurosynth/neurosynth>), which represented the association between brain voxels and terms of neurocognitive processes as the probability that a given term was reported in a brain-imaging experiment if neural activation was observed at a given voxel. Specifically, a total of 123 activation maps that were primarily related to cognitive function were selected based on the Cognitive Atlas (Poldrack et al., 2011). The full list of these maps can be found in Hansen et al. (2022). The activation maps were parcellated using the Schaefer's atlas and Z-transformed to obtain regional activation scores, which were used to predict nodal degrees of the CBF networks. Again, only the PLS1 was examined, whose significance level was estimated via a spin-based permutation test (10,000 times) (Alexander-Bloch et al., 2018; Váša et al., 2018). If the correlation was significant, we further computed Pearson correlation between the PLS1 scores and regional activation scores for each activation map. Significance levels of these correlations were estimated using spin-based permutation tests (10,000 times), followed by multiple comparison correction across all correlations using the Bonferroni method ($p < 0.05/123$).

3. Results

3.1. Connectivity and topological organization of CBF networks

Connectivity patterns. Fig. 2a shows the CBF similarity matrix averaged across all subjects and scans. High similarity was observed between brain regions (0.708 ± 0.096).

Global organization. Each CBF network exhibited small-worldness, high efficiency, and modular architecture over the majority of the thresholding range, as evidenced by normalized $C_p > 1$ and normalized $L_p \sim 1$, normalized $E_{loc} > 1$ and normalized $E_{glob} \sim 1$, and normalized $Q > 1$ (Fig. 2b).

Hubs. Degree distribution of the CBF networks was best fitted by the exponentially truncated power law model ($R^2 = 0.985$) (Fig. 2c). A total of 20 regions were identified as hubs that were predominantly located in the DAN (5), DMN (5), SMN (4), and VAN (4) (Fig. 2d).

3.2. Reliability and consistency of CBF networks

Global network measures of the CBF networks exhibited low TRT reliability (C_p : ICC = 0.347; L_p : ICC = 0.200; E_{loc} : ICC = 0.345; E_{glob} : ICC = 0.277; Q : ICC = 0.259; normalized C_p : ICC = 0.222; normalized L_p : ICC = 0.249; normalized E_{loc} : ICC = 0.221; normalized E_{glob} : ICC = 0.203; normalized Q : ICC = 0.314). For nodal degrees, although the overall TRT reliability was low (ICC = 0.210 ± 0.104), a specific set of regions (9, 4.5%) showed fair TRT reliability (Fig. 3a). These regions were predominantly located in the VAN (4, 44.4%) and DMN (3, 33.3%), such as the right temporal-occipital-parietal junction region and left prefrontal cortex. Similarly, despite an overall low TRT reliability for interregional CBF similarities (ICC = 0.206 ± 0.102), a specific set of edges (888, 4.46%) showed fair TRT reliability (Fig. 3b). For between-subject consistency, both nodal degrees ($r = 0.746 \pm 0.103$) (Fig. 3c) and interregional CBF similarities ($r = 0.654 \pm 0.112$) (Fig. 3d) showed high consistency.

3.3. Inter-subnetwork differences of CBF networks

The high-order subnetworks exhibited significantly higher intra-network CBF similarities than the low-order subnetworks ($t = 7.740$, $p < 0.001$) (Fig. 4a). Moreover, compared to the low-order subnetworks, the high-order subnetworks showed significantly lower TRT reliability for intra-subnetwork CBF similarities ($t = -9.210$, $p < 0.001$) (Fig. 4b). No significant differences were observed for neither nodal degrees ($t = 0.860$, $p = 0.383$) nor the TRT reliability of nodal degrees ($t = 1.259$, $p = 0.206$) between the low- and high-order subnetworks.

3.4. Influencing factors of CBF networks

Nodal degrees of the CBF networks were significantly negatively correlated with regional sizes ($\rho = -0.339$, $p < 0.001$) (Fig. 5a) but positively correlated with regional mean CBF levels ($\rho = 0.601$, $p < 0.001$) (Fig. 5b). For interregional CBF similarities, a significantly negative correlation was correlated with interregional anatomical

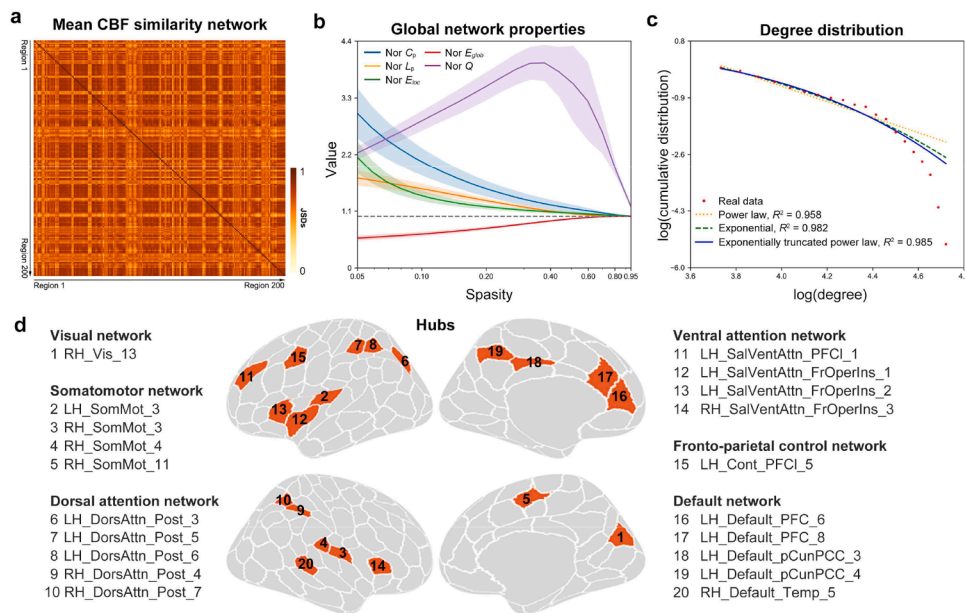


Fig. 2. Topological organization of the CBF networks. (a) Group-level mean CBF similarity matrix. (b) The CBF networks showed small-world organization, high parallel efficiency, and modularity. (c) The degree distribution of the CBF networks was best fitted by the exponentially truncated power law model. (d) A total of 20 regions were identified as hubs in the CBF networks. CBF, cerebral blood flow; JSDs, Jensen-Shannon divergence-based similarity; Nor C_p , normalized clustering coefficient; Nor L_p , normalized shortest path length; Nor E_{loc} , normalized local efficiency; Nor E_{glob} , normalized global efficiency; Nor Q , normalized modularity; LH, left hemisphere; RH, right hemisphere; FrOper, frontal operculum; Ins, insula; PFC1, lateral prefrontal cortex; pCunPCC, precuneus posterior cingulate cortex; Temp, temporal.

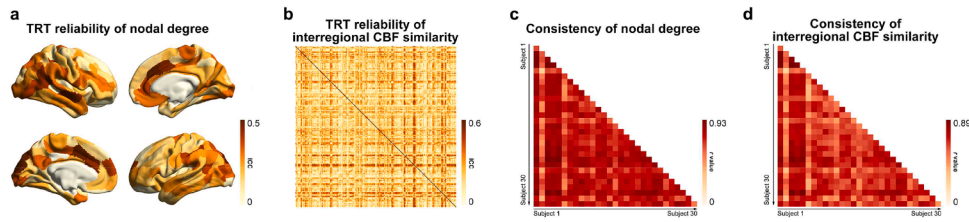


Fig. 3. The test-retest reliability and between-subject consistency of the CBF networks. The CBF networks exhibited low-to-fair test-retest reliability and high between-subject consistency for both nodal degrees (a and c) and interregional CBF similarities (b and d). TRT, test-retest; CBF, cerebral blood flow; ICC, intraclass correlation coefficient.

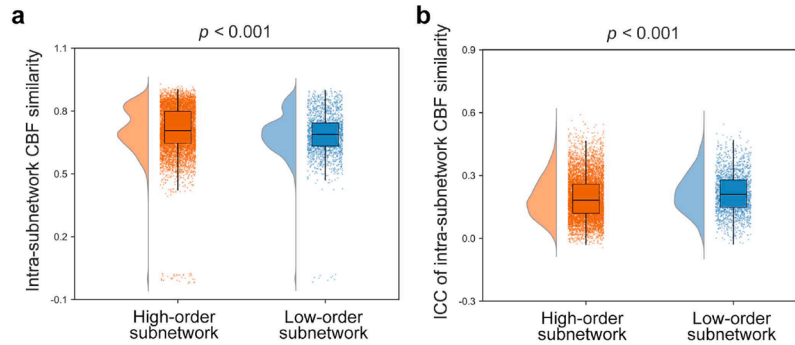


Fig. 4. The difference between low- and high-order subnetworks in (a) interregional CBF similarities and (b) their TRT reliability. Compared with low-order subnetworks, high-order subnetworks showed significantly higher interregional CBF similarities but lower TRT reliability of these similarities. CBF, cerebral blood flow.

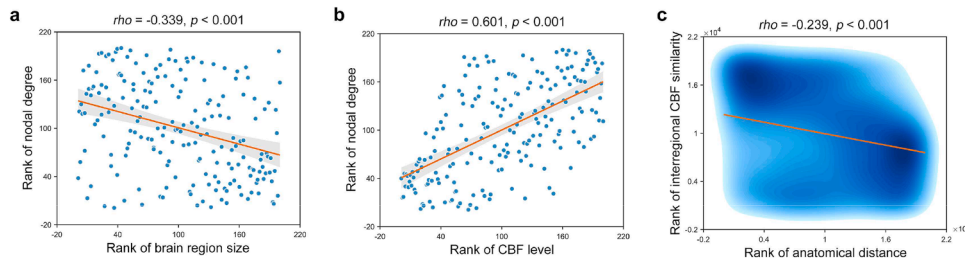


Fig. 5. Factors affecting the CBF networks. (a) Nodal degrees of the CBF networks showed a significantly negative correlation with regional sizes. (b) Nodal degrees of the CBF networks showed a significantly positive correlation with regional CBF levels. (c) Interregional CBF similarities exhibited a significantly negative correlation interregional anatomical distances. All correlations were based on nonparametric Spearman correlation and thus the ranks of the corresponding variables were displayed. CBF, cerebral blood flow.

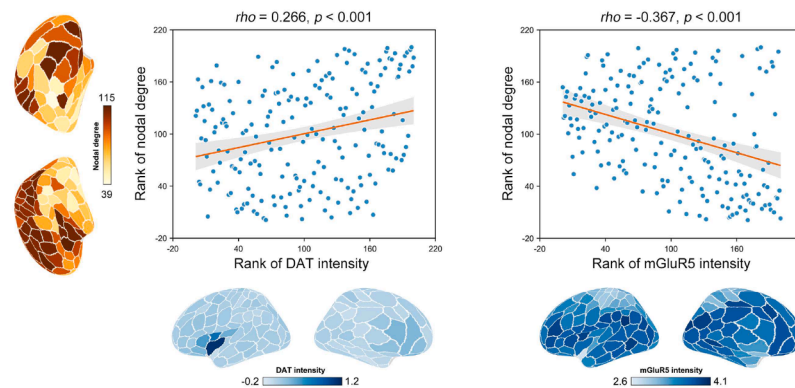


Fig. 6. Chemoarchitectonic correlates of the CBF networks. Nodal degrees of the CBF networks showed a significantly positive correlation with regional intensities of DAT and a significantly negative correlation with regional intensities of mGluR5. All correlations were based on nonparametric Spearman correlation and thus the ranks of the corresponding variables were displayed. DAT, dopamine transporter; mGluR5, metabolic glutamate receptor 5.

distances ($\rho = -0.239, p < 0.001$) (Fig. 5c).

3.5. Chemoarchitectonic correlates of CBF networks

Nodal degrees of the CBF networks exhibited a significantly positive correlation with the intensities of DAT with [^{11}C]P943 tracer ($\rho = 0.266, p < 0.001$) and a significantly negative correlation with the intensities of mGluR5 with [^{11}C]ABP688 tracer ($\rho = -0.367, p = 0.002$) (Fig. 6). No significant correlations were observed for the intensities of other neurotransmitter maps ($p > 0.05$, FDR corrected).

3.6. Genetic correlates of CBF networks

A significantly positive correlation was observed between nodal degrees of the CBF networks and the PLS1 scores derived from regional gene expression profiles ($r = 0.530, p = 0.043$) (Fig. 7a). A total of 427 and 383 genes were identified that positively and negatively contributed to this correlation, respectively (Fig. 7b). GO enrichment analysis revealed that the PLS1+ genes were enriched in 17 biological processes, which were primarily involved in cholesterol-related processes (e.g., ‘intracellular cholesterol transport’ and ‘cholesterol homeostasis’) and endothelial cell-related processes (e.g., ‘positive regulation of endothelial cells apoptotic process’ and ‘regulation of endothelial cells differentiation’) (Table S2). No neurobiological processes were observed for the PLS1- genes. Cell type-specific analysis revealed a significantly positive correlation between nodal degrees of the CBF networks and the mean expression levels of genes enriched in the endothelial cells ($\rho = 0.332, p < 0.001$) (Fig. 7c). No significant correlations were observed for other types of cell types ($p > 0.05$, FDR corrected).

3.7. Cognitive correlates of CBF networks

We observed a significantly positive correlation between nodal degrees of the CBF networks and the PLS1 scores derived from the 123 meta-analytic activation maps ($r = 0.573, p = 0.003$) (Fig. 8a). The PLS1 scores were significantly positively correlated with 24 and negatively correlated with 28 activation maps ($p < 0.05$, Bonferroni corrected). The activation maps showing the highest positive correlations were mainly related to executive function (e.g., ‘goal’, ‘monitoring’, ‘interference’, ‘response selection’, and ‘planning’) and ‘working memory’ while the activation maps with the highest negative correlations were primarily associated with the language domain, such as ‘communication’, ‘meaning’, ‘listening’, ‘language comprehension’, ‘speech perception’, and ‘language’ (Fig. 8b).

4. Discussion

In this study, we introduced an approach to construct functional brain networks by estimating shape similarity in probability distributions of regional static CBF. We systematically evaluated the constructed CBF networks by analyzing their topological organization, TRT

reliability, between-subject consistency, alignment with cortical functional topography, influence factors, and associations with chemoarchitecture, gene expression, and cognition. We found that the CBF networks exhibited non-trivial organization, low-to-fair TRT reliability, and high between-subject consistency. We further demonstrated that the CBF networks were related to regional size and CBF level, depended on anatomical distance, and were modulated by cortical functional topography. Finally, we showed that the CBF networks covaried with maps of DAT and mGluR5 intensities, were associated with genes enriched in cholesterol- and endothelial cell-related processes, and encoded language, memory, and executive function. Altogether, these findings suggest that shape similarity in probability distributions of static CBF between regions offers a novel, reliable, and neurobiologically meaningful approach for the investigation of functional brain networks, serving as an important complement to current methods.

4.1. Nontrivial organization of CBF networks

We found that the CBF networks exhibited small-world organization that is thought to support efficient segregation and integration of information in the human brain while minimizing wiring cost (Bassett and Bullmore, 2006; Liao et al., 2017). The small-world organization has been frequently observed in other types of brain networks, such as BOLD-based functional networks (Wang et al., 2009a), structural networks (Hagmann et al., 2007), and morphological networks (Li et al., 2021a; Wang et al., 2016). Consistent with previous studies, our findings provide further evidence for the idea that the small-world organization is a universal organizational principle of the human brain networks. In addition to the small-world organization, we found that the CBF network exhibited high modularity, indicating that the CBF networks can be divided into several distinct, interacting modules. Modularity allows efficient information processing in a network by localizing certain functions to specific modules while maintaining inter-module communications. Thus, modularity may improve the robustness, adaptivity, and evolvability of the human brain networks and support the emergence of adaptive behavior and cognition (Bullmore and Sporns, 2009; Meunier et al., 2010). Finally, we identified several hubs in the CBF networks and found a significantly positive correlation between nodal degrees of the CBF networks and regional CBF levels. These findings suggest that regions with high CBF levels tend to be hubs of the CBF networks, consistent with results from BOLD-based functional brain networks (Liang et al., 2013b). Previous studies have shown that regions with high CBF levels are typically associated with enhanced neuronal activities, dense capillary networks, and increased energy demands for oxygen and glucose (Karbowski, 2011; Watts et al., 2018). These features may make them engage in more neural processing and thus become hubs in functional brain networks.

4.2. TRT reliability of CBF networks

In this study, we evaluated the TRT reliability and between-subject

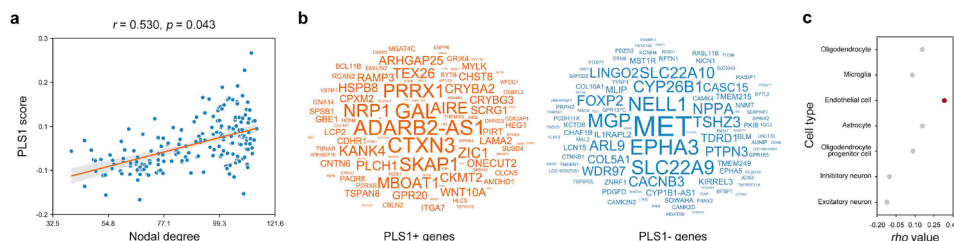


Fig. 7. Genetic correlates of the CBF networks. (a) Nodal degrees of the CBF networks showed a significantly positive correlation with the PLS1 scores derived from transcriptional activity of 15,633 genes. (b) A total of 427 and 383 genes were identified to show positive and negative contribution to the PLS1 scores, respectively. (c) Nodal degrees of the CBF networks showed a significantly positive correlation with the mean transcriptional activity of genes enriched in the endothelial cell. PLS, partial least-squares; FDR, false discovery rate.

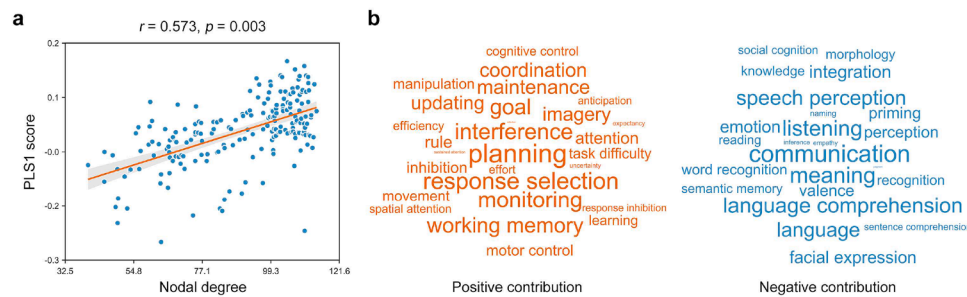


Fig. 8. Cognitive relevance of the CBF networks. **(a)** Nodal degrees of the CBF networks showed a significantly positive correlation with the PLS1 scores derived from 123 activation maps. **(b)** A total of 24 and 28 cognitive processes were identified to show positive and negative contribution to the PLS1 scores, respectively. PLS, partial least-squares.

consistency of the CBF networks. We observed low-to-fair reliability for global network measures, nodal degrees, and interregional CBF similarities. Previous studies have shown that CBF is affected by physical and emotional states of participants (Honda et al., 2018; Hoshi and Chen, 2002; Ozawa et al., 2019). Furthermore, CBF is found to vary considerably with diurnal variations (Hodkinson et al., 2014; Shannon et al., 2013) and across the menstrual cycle in females (Otomo et al., 2020). However, these factors were not taken into account during the data collection phase, which are speculated to collectively result in, to some extent, the observed low-to-fair TRT reliability of the CBF networks. In future studies, it would be interesting to investigate the extent to which the CBF networks constructed with our proposed approach are affected by these factors. In addition to these factors, different image preprocessing strategies (e.g., motion correction and spatial filtering) are previously reported to influence the TRT reliability of CBF (Fazlollahi et al., 2015), which may also contribute to the observed low-to-fair TRT reliability of the CBF networks. Parsing the effects of different image preprocessing strategies on the TRT reliability of the CBF networks will provide guidance on determining optimized analytical strategies for obtaining reliable CBF networks. It should be noted that despite the low-to-fair TRT reliability, we observed high consistency across subjects for the CBF networks. The high consistency suggests the presence of a stable, intrinsic CBF network patterning shared across subjects. Thus, we averaged individual CBF networks to derive a consensus estimate, which was used to explore neurobiological signatures of the CBF networks.

4.3. Neurobiological signatures of CBF networks

We examined neurobiological signatures of the CBF networks by linking them with neurotransmitter intensities and transcriptional profiles. We observed a significantly positive correlation between nodal degrees of the CBF networks and DAT intensities negative correlation with mGluR5 intensities. A previous study demonstrated that CBF changes induced by different psychiatric medications were associated with neurotransmitter systems, including DAT (Dukart et al., 2018). Regarding mGluR5, its activation was found to trigger Ca^{2+} transients in astrocytes (Wang et al., 2006), which initiated a series of events that ultimately affected CBF through the regulation of arteriole smooth muscle tone (Zonta et al., 2003). Thus, the observed associations of the CBF networks with DAT and mGluR5 intensities might presumably be driven by the modulatory effects of these neurotransmitters on CBF dynamics.

In addition to the chemoarchitectonic correlates, we showed that the CBF networks were significantly correlated with transcriptional profiles. Furthermore, we found that genes showing strong positive contribution to this correlation were primarily enriched in cholesterol-related processes, including intracellular cholesterol transport and cholesterol homeostasis. Physiologically, the cholesterol play a critical role in maintaining the optimal level of energetic metabolism (Czuba et al., 2017). Considering that CBF is closely coupled with glucose and oxygen

metabolism (Hoge et al., 1999; Paulson et al., 2010; Vaishnavi et al., 2010b), we speculate that the cholesterol modulates energetic demands and therefore CBF distribution in the brain, which further affects topological organization of the CBF networks. This speculation sounds plausible given our observed positive correlation between regional CBF and nodal degrees of the CBF networks. In addition to the cholesterol-related processes, the PLS1+ genes were enriched in several endothelial cell-related processes, such as positive regulation of endothelial cells apoptotic process and regulation of endothelial cells differentiation. These findings is consistent with our cell type-specific analysis, which revealed a significantly positive correlation between nodal degrees of the CBF networks and mean expression levels of genes enriched in the endothelial cells. Several previous studies have reported the roles of endothelial cells in CBF regulation, including vascular tone regulation, inflammatory response, thrombosis, adhesion, and vascular permeability (Ashby and Mack, 2021; Cohen, 1995; Godo and Shimokawa, 2017). Here, our results suggest that the endothelial cells may also play a critical role in the topological organization of the CBF networks. Taken together, these findings provide important insights into understanding the neurobiological substrates of the CBF networks constructed with our proposed approach.

4.4. Cognitive and behavioral relevance of CBF networks

In this study, we demonstrated cognitive and behavioral relevance of the CBF networks by relating them with a series of meta-analytic activation maps. We found that the CBF networks were mainly related to executive function, working memory, and language processing. These findings are in line with the perspective that cognition results from functional interactions of distributed brain systems operating in large-scale networks (Bressler and Menon, 2010). Notably, a previous study found that higher regional CBF was associated with better performance in executive function and memory (Leeuwis et al., 2018). Thus, cognitive performance seems to be not only associated with regional CBF but also related to topological organization of the CBF networks. In future studies, it is warranted to study whether the CBF networks are able to account for interindividual variance and predict individual scores in behavior and cognition. In particular, given the high modularity of the CBF networks, it is interesting to identify modular compositions of the CBF networks and to determine how different modules are specifically related to diverse behavioral and cognitive functions.

4.5. Limitations and future directions

Several limitations exist in this study. First, there are inherent issues associated with ASL imaging technology, such as vascular artifacts. Although vascular crushing is used during the ASL data collection, we are not sure whether our results are still confounded by vascular artifacts. Future studies are thus needed to examine the robustness of our method by using ASL data with multiple label durations and/or post-

labeling delays, which are able to account for the effects of variable arterial transit time on accurate assessment of cerebral perfusion (Alsop et al., 2015; Woods et al., 2024). Second, we examined the biological meaning of the CBF networks by associating them with publicly available datasets. This may result in an underestimation of the chemo-architectonic and genetic correlates of the CBF networks. Further work is required to validate our results by collecting all data from the same cohort of participants. Third, there are numerous approaches to construct functional brain networks using CBF currently (Liang et al., 2014b; Liu et al., 2016). Thus, it would be important to compare the proposed approach in this study with these existing ones to better understand its pros and cons. Fourth, previous studies have demonstrated profound effects of different network analytical strategies on the topological organization of human brain networks (Brodoehl et al., 2020; Liang et al., 2012; Wang et al., 2009b; Zalesky et al., 2010). Therefore, future studies are needed to explore the reproducibility of our results when different choices of brain parcellation, thresholding method, and network type are used. Fifth, we utilized static CBF to construct functional brain networks since regional CBF is highly correlated with neuronal activity due to Neuro-Vascular coupling. However, a previous found that the dynamic range of the CBF response was smaller than that of the BOLD response in luminance contrast of a visual stimulus (Liang et al., 2013a). Thus, compared with BOLD-based functional connectivity approach, our proposed approach may be less sensitive in revealing the reorganization of functional brain networks evoked by external stimuli. Future studies are thus warranted to examine similarities and differences between functional brain networks derived from these two types of approaches. Finally, alterations in static CBF have been found in several neuropsychiatric disorders (Falcon et al., 2024; Li et al., 2021a; Percie Du Sert et al., 2023). Accordingly, an interesting topic in the future is to investigate whether CBF networks constructed with our proposed approach are also disrupted in these disorders.

Data and code availability statement

All data that support the findings of this study are from publicly available datasets (CBF dataset: https://fcon_1000.projects.nitrc.org/indi/CoRR/html/hnu_1.html; Neurotransmitter receptor and transporter dataset: <https://github.com/netneurolab/neuromaps>; Meta-analytic activation dataset: <https://neurosynth.org/>). Code for all analyses in this study are from publicly available toolboxes, including GREYNA (<https://www.nitrc.org/projects/gretna>; topological analysis), abagen (version 0.1.3; <https://github.com/rmarkello/abagen>; preprocess for gene data), and GeneCategoryEnrichmentAnalysis (<https://github.com/benfulcher/GeneCategoryEnrichmentAnalysis>; gene ontology enrichment analysis).

CRedit authorship contribution statement

Changwen Wu: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Yu He:** Formal analysis. **Junle Li:** Writing – review & editing, Formal analysis. **Xiaofan Qiu:** Writing – review & editing, Formal analysis. **Qihong Zou:** Writing – review & editing. **Jinhui Wang:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.neuroimage.2025.121069](https://doi.org/10.1016/j.neuroimage.2025.121069).

Data availability

All data that support the findings of this study are from publicly available datasets. Code for all analyses in this study are from publicly available toolboxes.

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